

# An introduction to Bayesian modelling with brms and Stan

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<https://frodriguezsanchez.net>

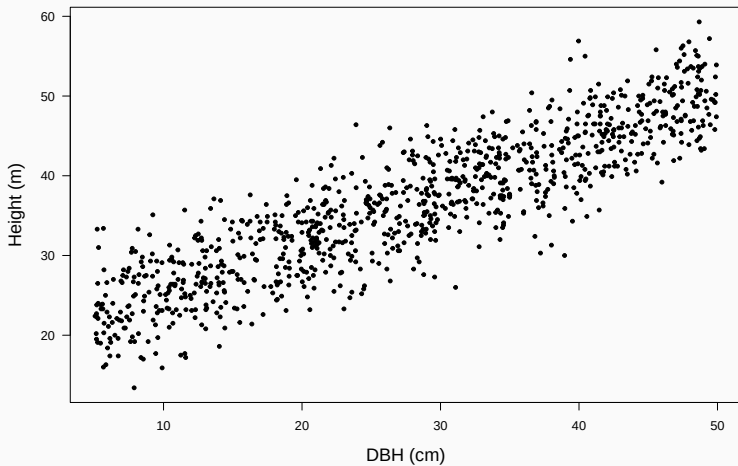
## Our dataset: tree heights and DBH

- One species
- 10 plots
- 1000 trees
- Number of trees per plot ranging from 4 to 392

```
trees <- read.csv('data/trees.csv')
```

|          | site  | dbh           | height        |
|----------|-------|---------------|---------------|
| Min.     | : 1.0 | Min. : 5.06   | Min. :13.40   |
| 1st Qu.: | 1.0   | 1st Qu.:17.69 | 1st Qu.:29.68 |
| Median   | : 2.0 | Median :28.62 | Median :36.55 |
| Mean     | : 2.7 | Mean :27.88   | Mean :36.51   |
| 3rd Qu.: | 4.0   | 3rd Qu.:38.97 | 3rd Qu.:43.33 |
| Max.     | :10.0 | Max. :49.92   | Max. :59.30   |

## What's the relationship between DBH and height?



## First step: linear regression (lm)

```
simple.lm <- lm(height ~ dbh, data = trees)
```

Call:

```
lm(formula = height ~ dbh, data = trees)
```

Residuals:

| Min      | 1Q      | Median | 3Q     | Max     |
|----------|---------|--------|--------|---------|
| -13.3270 | -2.8978 | 0.1057 | 2.7924 | 12.9511 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )   |
|-------------|----------|------------|---------|------------|
| (Intercept) | 19.33920 | 0.31064    | 62.26   | <2e-16 *** |
| dbh         | 0.61570  | 0.01013    | 60.79   | <2e-16 *** |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.093 on 998 degrees of freedom

Multiple R-squared: 0.7874, Adjusted R-squared: 0.7871

F-statistic: 3695 on 1 and 998 DF, p-value: < 2.2e-16

## Center continuous variables

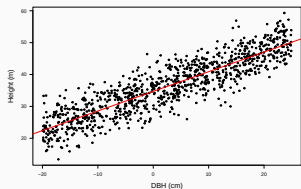
```
summary(trees$dbh)
```

| Min. | 1st Qu. | Median | Mean  | 3rd Qu. | Max.  |
|------|---------|--------|-------|---------|-------|
| 5.06 | 17.69   | 28.62  | 27.88 | 38.97   | 49.92 |

```
trees$dbh.c <- trees$dbh - 25
```

So, all parameters will be referred to a 25 cm DBH tree.

# Linear regression with centred DBH



```
lm(formula = height ~ dbh.c, data = tree)
      coef.est coef.se
(Intercept)  34.73    0.13
dbh.c         0.62    0.01
---
n = 1000, k = 2
residual sd = 4.09, R-Squared = 0.79
```

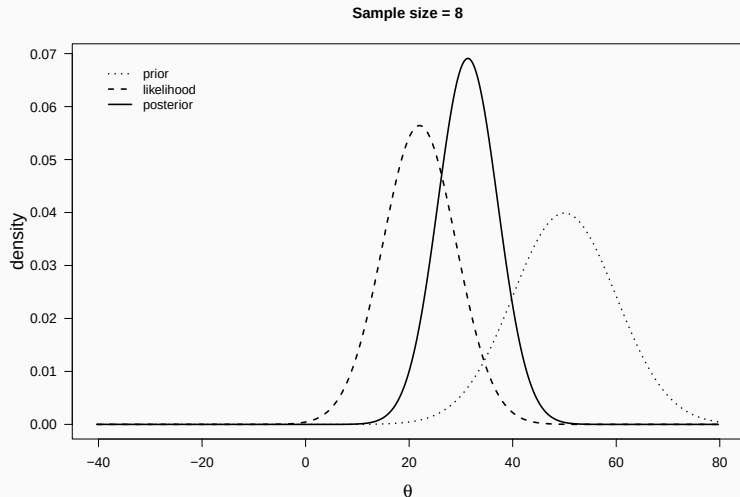
Let's make it Bayesian

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# Bayesian inference: prior, posterior, and likelihood

$$P(\text{Unknown}|\text{Data}) \propto P(\text{Data}|\text{Unknown}) \times P(\text{Unknown})$$

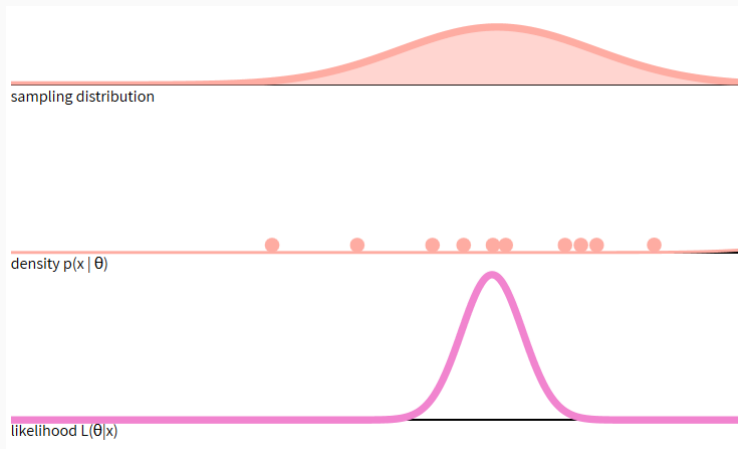
$$\text{Posterior} \propto \text{Likelihood} \times \text{Prior}$$





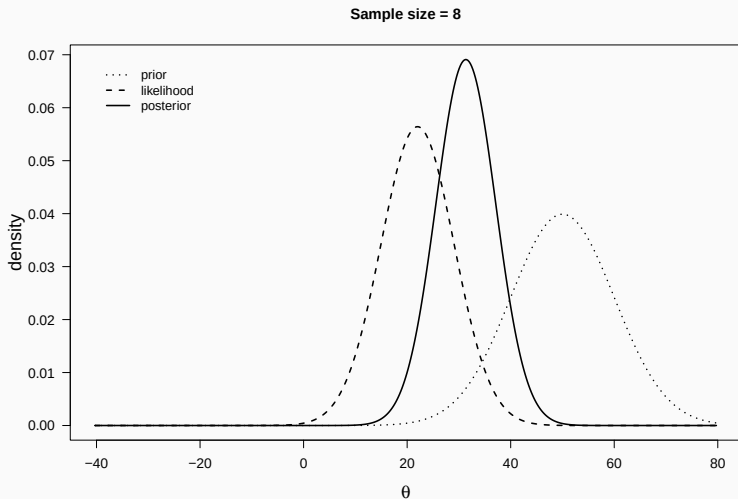
# What is the likelihood?

$$L(\theta|x) = P(x|\theta)$$

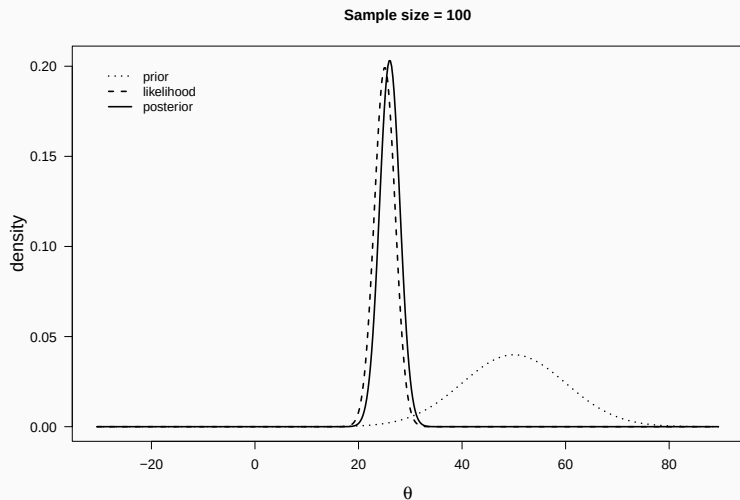


<https://seeing-theory.brown.edu/bayesian-inference/index.html>

# Bayesian inference: prior and likelihood produce posterior



## With increasing sample size, likelihood dominates prior



- Integrate information (**prior**)

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- Large dataset -> prior effect **diminishes**
- **Uncertainty / Propagate errors**



$$y_i \sim N(\mu_i, \sigma^2)$$

$$\mu_i = \alpha + \beta x_i$$

In this case:

$$\text{Height}_i \sim N(\mu_i, \sigma^2)$$

$$\mu_i = \alpha + \beta \text{DBH}_i$$

$\alpha$ : expected height when DBH = 25 cm

$\beta$ : how much height increases with every unit increase of DBH

```
library('brms')  
  
height.formu <- brmsformula(height ~ dbh.c)
```

We must define **prior distributions** for every parameter

```
get_prior(height.formu, data = trees)
```

|                          | prior     | class | coef  | group | resp | dpar | nlpar | lb | ub | tag |
|--------------------------|-----------|-------|-------|-------|------|------|-------|----|----|-----|
|                          | (flat)    | b     |       |       |      |      |       |    |    |     |
|                          | (flat)    | b     | dbh.c |       |      |      |       |    |    |     |
| student_t(3, 36.5, 10.2) | Intercept |       |       |       |      |      |       |    |    |     |
| student_t(3, 0, 10.2)    | sigma     |       |       |       |      |      |       | 0  |    |     |
| source                   |           |       |       |       |      |      |       |    |    |     |
| default                  |           |       |       |       |      |      |       |    |    |     |
| (vectorized)             |           |       |       |       |      |      |       |    |    |     |
| default                  |           |       |       |       |      |      |       |    |    |     |
| default                  |           |       |       |       |      |      |       |    |    |     |

# Choosing priors

Avoid 'non-informative' priors

Use *weakly informative* (e.g. relatively wide Normal or t-student distributions)  
or *strongly informative* priors based on previous knowledge and common sense.

Some tips for setting priors:

- <https://github.com/stan-dev/stan/wiki/Prior-Choice-Recommendations>

Run **prior predictive checks** (just priors, no data)

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- <https://doi.org/10.1111/oik.05985>

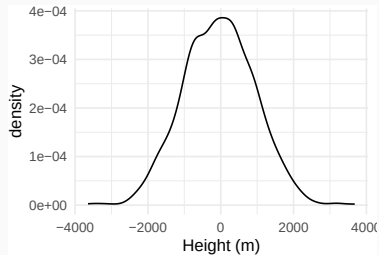
Run **prior predictive checks** (just priors, no data)

<https://distribution-explorer.github.io/>

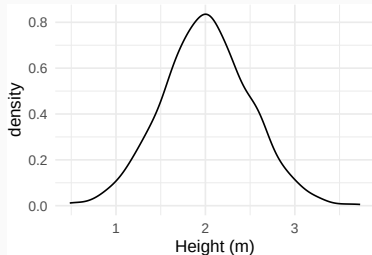


## Example: estimating people height across countries

Unreasonable prior



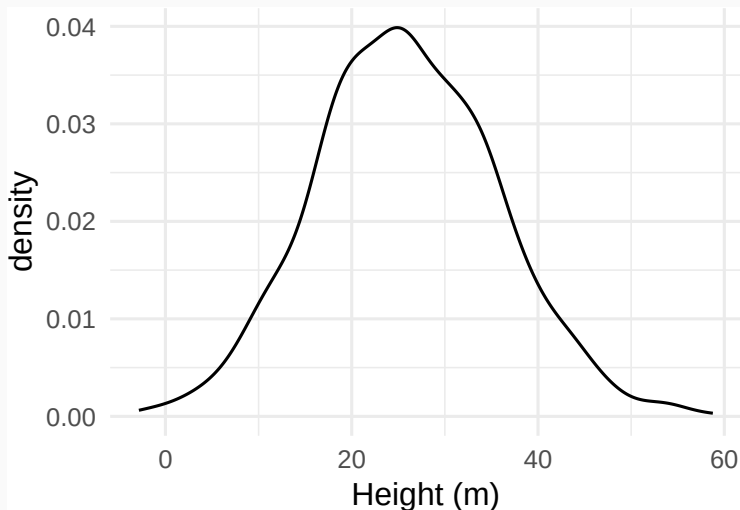
Reasonable prior



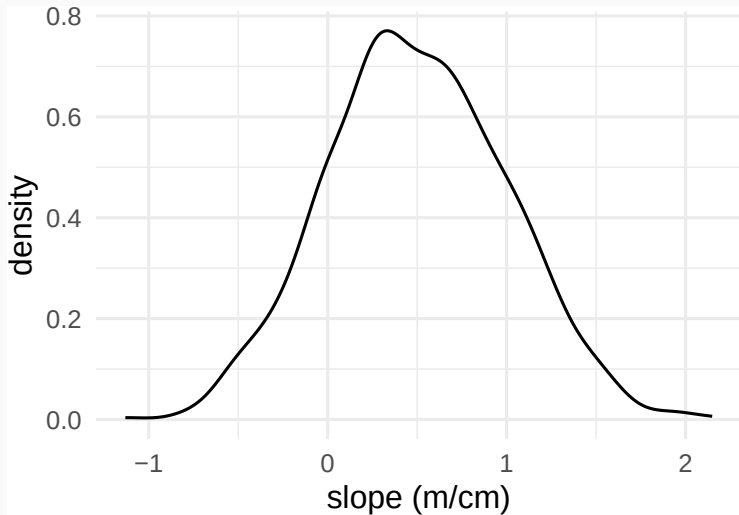
## Defining priors for our trees example

```
priors <- c(  
  set_prior('normal(30, 10)', class = 'Intercept'),  
  set_prior('normal(0.5, 0.4)', class = 'b'),  
  set_prior('normal(0, 5)', class = 'sigma')  
)
```

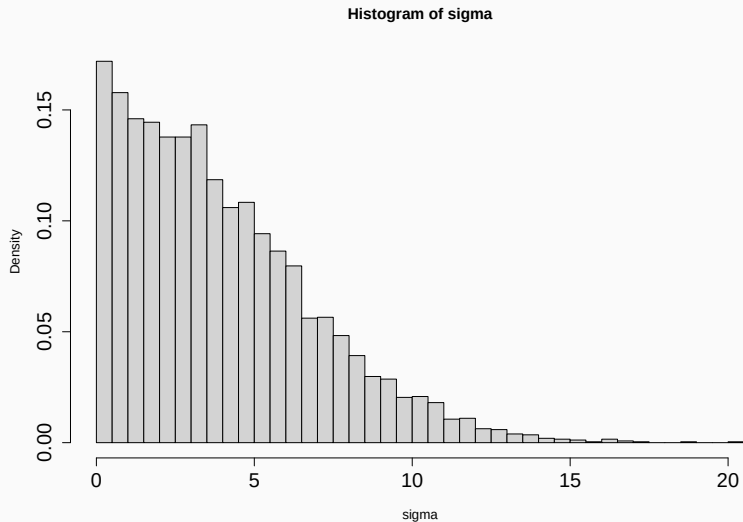
## Prior for intercept (average height of 25-cm diameter tree)



## Prior for slope



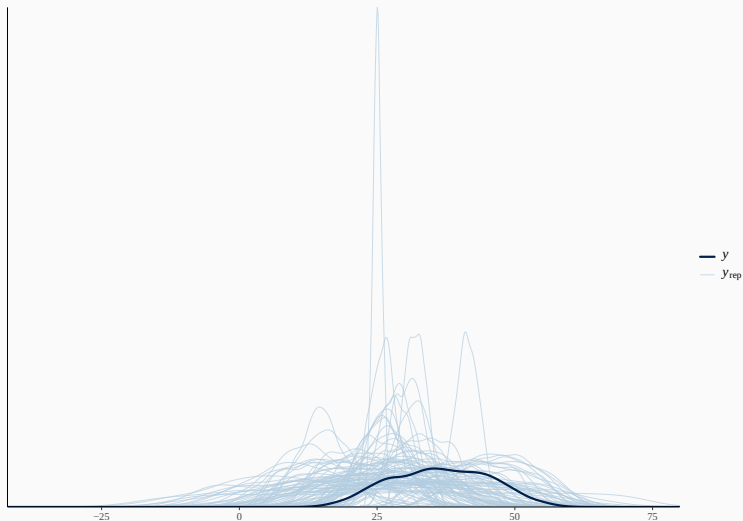
## Prior for sigma (residual sd)



```
height.mod <- brm(height.formu,  
  data = trees,  
  prior = priors,  
  sample_prior = 'only')
```

## Prior predictive check

```
pp_check(height.mod, ndraws = 100)
```



```
height.mod <- brm(height.formu,  
  data = trees,  
  prior = priors)
```



# Model summary

```
summary(height.mod)
```

```
Family: gaussian
Links: mu = identity
Formula: height ~ dbh.c
Data: trees (Number of observations: 1000)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
       total post-warmup draws = 4000
```

## Regression Coefficients:

|           | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|-----------|----------|-----------|----------|----------|------|----------|----------|
| Intercept | 34.73    | 0.13      | 34.47    | 34.99    | 1.00 | 4222     | 3247     |
| dbh.c     | 0.62     | 0.01      | 0.60     | 0.64     | 1.00 | 4499     | 2673     |

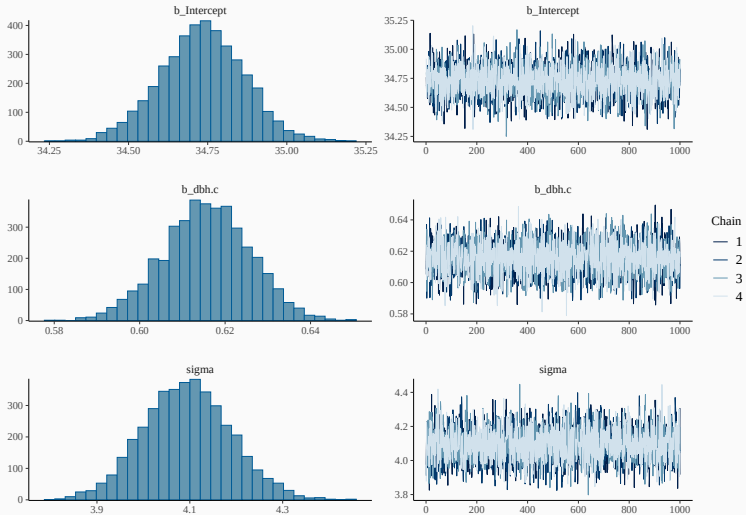
## Further Distributional Parameters:

|       | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|-------|----------|-----------|----------|----------|------|----------|----------|
| sigma | 4.09     | 0.09      | 3.92     | 4.28     | 1.00 | 4430     | 3153     |

Draws were sampled using sampling(NUTS). For each parameter, Bulk\_ESS and Tail\_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

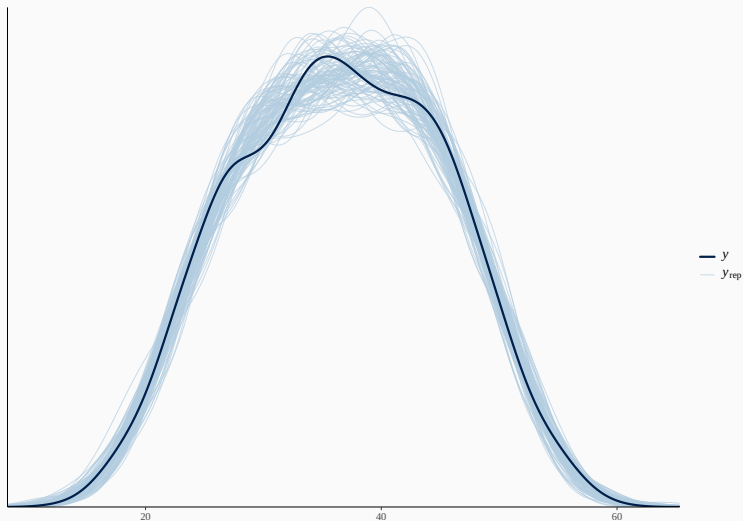
# Model visualisation

```
plot(height.mod)
```



# Posterior predictive checking

```
pp_check(height.mod, ndraws = 100)
```



```
library('shinystan')  
launch_shinystan(height.mod)
```

Sensitivity based on cjs\_dist

Prior selection: all priors

Likelihood selection: all data

| variable    | prior | likelihood | diagnosis |
|-------------|-------|------------|-----------|
| b_Intercept | 0.001 | 0.084      | -         |
| b_dbh.c     | 0.001 | 0.087      | -         |
| sigma       | 0.003 | 0.100      | -         |
| Intercept   | 0.001 | 0.085      | -         |

# Prior sensitivity analysis

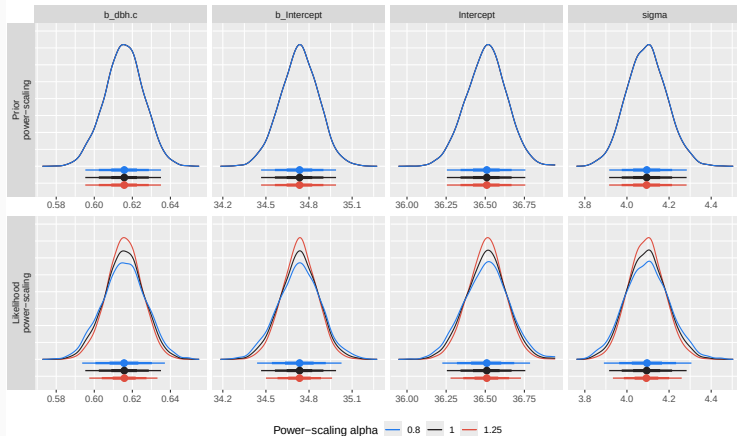
## Power-scaling sensitivity

Posterior density estimates depending on amount of power-scaling ( $\alpha$ ).

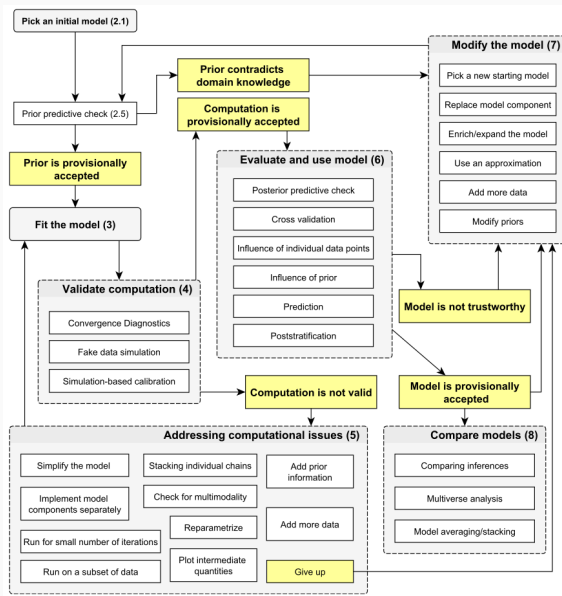
Overlapping lines indicate low sensitivity.

Wider gaps between lines indicate greater sensitivity.

Estimates with high Pareto  $k$  (dashed lines) may be inaccurate.



# The Bayesian workflow



height ~ sex



[Regression and other stories](#)

[Statistical Rethinking](#)

[Statistical rethinking with brms, ggplot2, and the tidyverse](#)

[Bayesian Population Analysis using WinBugs](#)

[Applied Hierarchical Modeling in Ecology](#)

[Stan user guide](#)